4 **(a)** State, in words, the relation between the electric field strength E and potential V at a point.

.....

..... [1]

(b) Two positively charged metal spheres, P and Q, of diameters 32 cm and 16 cm respectively, each carrying a charge of +7.2 nC, are isolated in space, as shown in Fig. 4.1.

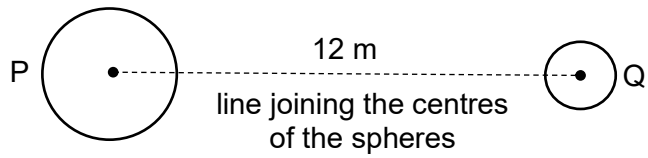

Fig. 4.1

The centres of the spheres are separated by a distance of 12 m. The distance x is measured from the centre of sphere P along the line joining the centres of the two spheres. Assume charges remain uniformly distributed on the surfaces of the spheres.

(i) State the value of x for which a stationary charged particle remains stationary when placed at this distance from the centre of sphere P.

x = ... m [1]

(ii) Calculate the electric potential at the point where the stationary charged particle remains stationary as stated in **(b)(i)**.

electric potential = ... V [1]

(iii) Sketch on Fig. 4.2, the variation with distance x of the electric potential V along the line joining the centres of the two spheres. Indicate on the horizontal axis your value of x in **(b)(i)**.

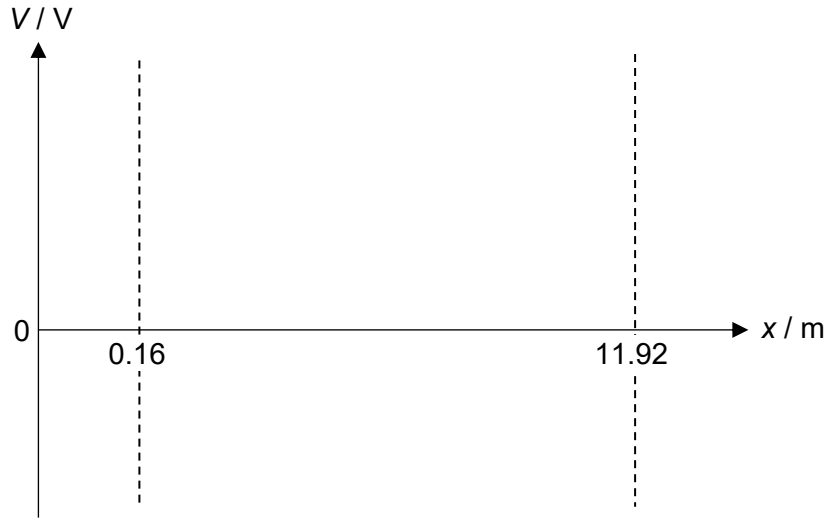

Fig. 4.2 (not to scale)

[3]

(iv) A positively charged particle is released at x = 0.16 m (surface of P).

With reference to your graph in **(b)(iii)**,

1. state and explain whether it will reach x = 11.92 m (surface of Q),

.....

.....

.....

..... [2]

2. describe and explain the entire motion of this particle, using energy considerations or otherwise.

.....

.....

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.....

.....

..... [3]